

Product Requirements Document

Digital Document Approval System

1. Product Summary

The Digital Document Approval System enables organizations to route documents through structured review, approval, signature, retention, and audit workflows. The system supports internal approvals, external signing, version tracking, approval status visibility, and compliance-ready audit history.

The product is intended for organizations that need controlled document workflows for contracts, policies, purchase approvals, HR documents, legal documents, compliance attestations, vendor documents, and operational forms.

2. Product Goals

The product should:

1. Allow users to submit documents for review and approval.
2. Route documents through configurable approval workflows.
3. Support sequential and parallel approvers.
4. Capture approvals, rejections, comments, and requested changes.
5. Integrate with an external e-signature provider where legally binding signatures are required.
6. Maintain complete document version history.
7. Maintain a tamper-evident audit trail of document lifecycle events.
8. Provide visibility into approval status, blockers, overdue tasks, and completed approvals.
9. Support document retention, archival, and access policies.

3. Non-Goals

The product will not:

1. Replace a full enterprise content management platform.
2. Provide advanced collaborative document editing.
3. Replace legal contract lifecycle management in the initial release.
4. Provide identity verification beyond integrations with configured identity or signature providers.
5. Provide payment processing.
6. Provide document generation from complex business templates in the initial release.

7. Act as the legal authority for signature validity beyond records received from the e-signature provider.
8. Replace organization-wide records management policy ownership.

4. Target Users

User	Description
Document Submitter	Uploads or creates a document and starts an approval workflow.
Approver	Reviews, approves, rejects, or requests changes to a document.
Reviewer	Provides comments or review feedback without final approval authority.
Workflow Administrator	Configures approval templates, roles, routing rules, escalation rules, and retention policies.
Records Manager	Oversees retention, archival, legal hold, access, and audit requirements.
External Signer	Signs documents through an external signature workflow.
Compliance Auditor	Reviews approval evidence, access history, signature status, and retention records.

5. External Systems

External System	Description
Identity Provider	Authenticates internal users and provides group or role information.
E-Signature Provider	Manages external or legally binding signature ceremonies.
Notification Provider	Sends email, SMS, or messaging notifications.
Document Storage Provider	Optional storage backend for document binaries and archives.
Enterprise Document Repository	Optional future integration for publishing approved documents or importing source documents.

6. Primary User Problems

Organizations often manage approvals through email threads, shared drives, chat messages, and manual status tracking. This makes it difficult to determine which version

was approved, who approved it, whether required signatures were completed, and whether the document was retained according to policy.

Submitters need a clear way to start and monitor approvals. Approvers need a simple review experience. Administrators need configurable routing rules. Records managers need reliable audit and retention controls. Auditors need defensible evidence of approval, signature, access, and retention activity.

7. Product Capabilities

7.1 Document Submission

Submitters should be able to:

1. Upload a document.
2. Enter required document metadata.
3. Select a document type.
4. Choose or trigger an approval workflow.
5. Add optional reviewers or approvers when permitted.
6. Submit the document for approval.
7. Track approval status.
8. Withdraw a submission before final approval where permitted.

Document metadata may include:

1. Title.
2. Document type.
3. Department.
4. Business owner.
5. Effective date.
6. Expiration date.
7. Confidentiality level.
8. Related project, vendor, employee, customer, or policy reference.
9. Retention category.
10. Signature requirement.
11. Approval due date.

7.2 Workflow Configuration

Workflow Administrators should be able to configure approval workflows based on:

1. Document type.
2. Department.
3. Dollar amount or risk level where applicable.
4. Confidentiality level.
5. Business owner.

6. Required approval roles.
7. Sequential approval stages.
8. Parallel approval stages.
9. Conditional routing rules.
10. Escalation timing.
11. Required signature rules.
12. Retention category.

The system should support reusable workflow templates.

7.3 Review and Approval

Approvers and reviewers should be able to:

1. View assigned documents.
2. Review document metadata.
3. Preview or download the document.
4. Add comments.
5. Approve the document.
6. Reject the document.
7. Request changes.
8. Delegate or reassign approval when permitted.
9. View prior approval history and comments.
10. See whether they are acting as a reviewer or final approver.

The system should preserve the approval decision, actor, timestamp, comment, and document version.

7.4 Version Management

The system should maintain document version history.

The system should:

1. Assign a unique document ID.
2. Assign a version number to each uploaded or revised document.
3. Prevent approval ambiguity between versions.
4. Require re-approval when a submitted document is materially changed.
5. Preserve historical approved versions.
6. Display the currently active version.
7. Track who uploaded or modified each version.
8. Preserve prior versions according to retention policy.

7.5 Approval Routing

The system should route approval tasks according to the active workflow.

Routing should support:

1. Sequential approval stages.
2. Parallel approval stages.
3. Conditional routing.
4. Role-based approver resolution.
5. Specific named approvers.
6. Reviewer-only steps.
7. Escalation on overdue tasks.
8. Rework loops when changes are requested.
9. Cancellation by authorized users.
10. Delegation or reassignment where configured.

7.6 External Signature

When a document requires external signature, the system should coordinate with an external e-signature provider.

The system should:

1. Prepare a signature request.
2. Send signer details and document package to the e-signature provider.
3. Track signature status.
4. Receive signature completion, decline, failure, or expiration events.
5. Store the signed document or provider reference.
6. Record signature events in audit history.
7. Continue or complete the approval workflow based on signature result.
8. Surface signature failures or stalled signature requests to authorized users.

The system should not bypass the e-signature provider for documents requiring legally binding signatures.

7.7 Notifications and Reminders

The system should notify users about:

1. New approval assignments.
2. Review requests.
3. Requested changes.
4. Approval completion.
5. Rejection.
6. Signature requests.
7. Overdue approval tasks.
8. Escalations.
9. Workflow cancellation.

10. Final document approval.
11. Retention or archival tasks where configured.

Notifications should include document title, task type, due date, and a secure link to the relevant task. Notifications should not expose sensitive document contents.

7.8 Status Tracking and Search

Users should be able to search and filter documents by:

1. Document title.
2. Document type.
3. Submitter.
4. Approver.
5. Department.
6. Status.
7. Due date.
8. Effective date.
9. Expiration date.
10. Confidentiality level.
11. Signature status.
12. Retention category.

The system should provide clear status indicators for submitted, in review, changes requested, rejected, approved, awaiting signature, signed, archived, expired, canceled, and on legal hold.

7.9 Audit Trail

The system should maintain an audit trail for significant document lifecycle events.

Audit events should include:

1. Document created.
2. Document version uploaded.
3. Metadata changed.
4. Workflow started.
5. Review task assigned.
6. Comment added.
7. Approval granted.
8. Rejection issued.
9. Changes requested.
10. Task reassigned or delegated.
11. Signature request created.

12. Signature completed, declined, failed, or expired.
13. Workflow completed.
14. Document archived.
15. Access policy changed.
16. Retention action performed.
17. Legal hold applied or removed.
18. Document accessed or downloaded where required by policy.

Audit records should include actor, timestamp, event type, document ID, version, and relevant context.

7.10 Retention and Archival

The system should support retention policies based on document type or retention category.

The system should:

1. Assign retention category to documents.
2. Calculate retention deadlines where configured.
3. Identify documents eligible for archival.
4. Restrict deletion to authorized roles.
5. Preserve audit history according to policy.
6. Support legal hold or retention override where required.
7. Provide retention status visibility to Records Managers.
8. Prevent archival or deletion actions that violate active legal hold.

8. Representative User Journeys

8.1 Submit Document for Approval

1. Submitter uploads a document.
2. Submitter enters required metadata.
3. System selects or recommends an approval workflow.
4. Submitter confirms submission.
5. System creates the document record and initial version.
6. System starts the workflow.
7. System assigns first approval tasks.
8. Approvers receive notifications.

8.2 Approve Document with Sequential Routing

1. Approver receives an approval task.
2. Approver opens the document.
3. Approver reviews metadata, content, and prior comments.
4. Approver approves the current version.

5. System records the approval decision.
6. System routes the document to the next approval stage.
7. Final approver approves the document.
8. System marks the document approved.
9. System notifies the submitter.

8.3 Request Changes

1. Approver reviews a document.
2. Approver identifies required changes.
3. Approver selects “request changes” and enters comments.
4. System records the decision.
5. System returns the workflow to the submitter.
6. Submitter uploads a revised version.
7. System creates a new document version.
8. System restarts the required approval stages.

8.4 Complete External Signature

1. Approval workflow reaches a signature-required stage.
2. System prepares the signature package.
3. System sends the request to the e-signature provider.
4. External signer receives the signature request.
5. External signer signs the document.
6. E-signature provider sends completion event.
7. System records signature completion.
8. System stores signed document or provider reference.
9. Workflow advances or completes.

8.5 Archive Approved Document

1. Document reaches approved or expired state.
2. Retention policy identifies archival eligibility.
3. Records Manager reviews retention status.
4. Records Manager archives the document.
5. System restricts normal editing.
6. System records archival action.
7. Document remains searchable according to access policy.

8.6 Auditor Reviews Approval Evidence

1. Compliance Auditor searches for an approved document.
2. System displays the approved version and metadata.
3. Auditor reviews workflow history.
4. Auditor reviews approver decisions and timestamps.
5. Auditor reviews signature status where applicable.

6. Auditor reviews retention and archival status.
7. Auditor exports or records evidence according to audit procedure.

9. Functional Requirements

ID	Requirement
FR-001	The system shall allow authorized users to upload documents.
FR-002	The system shall require configurable metadata for submitted documents.
FR-003	The system shall support reusable approval workflow templates.
FR-004	The system shall route documents through sequential approval stages.
FR-005	The system shall route documents through parallel approval stages where enabled.
FR-006	The system shall support conditional routing based on document metadata.
FR-007	The system shall allow approvers to approve, reject, or request changes.
FR-008	The system shall allow reviewers to comment without final approval authority.
FR-009	The system shall preserve approval decisions by document version.
FR-010	The system shall maintain document version history.
FR-011	The system shall prevent ambiguity about which document version was approved.
FR-012	The system shall restart required approval stages when a document version changes.
FR-013	The system shall integrate with an external e-signature provider where configured.
FR-014	The system shall track external signature status.
FR-015	The system shall store signed document output or signature provider reference.
FR-016	The system shall notify users about approval assignments, changes, overdue tasks, escalations, and completion events.
FR-017	The system shall provide document search

ID	Requirement
	and filtering.
FR-018	The system shall display document workflow status.
FR-019	The system shall support task reassignment or delegation where permitted.
FR-020	The system shall support escalation for overdue approval tasks.
FR-021	The system shall maintain audit history for document lifecycle events.
FR-022	The system shall enforce role-based access to documents, workflows, and administrative functions.
FR-023	The system shall support document retention categories and archival status.
FR-024	The system shall support legal hold or retention override where configured.
FR-025	The system shall expose integration failure status for notification, storage, identity, and e-signature workflows.
FR-026	The system shall allow authorized users to cancel an in-progress approval workflow.
FR-027	The system shall preserve approved document versions according to retention policy.

10. Non-Functional Requirements

Category	Requirement
Security	The system shall enforce authenticated and role-based access to documents, approvals, signatures, and administrative functions.
Confidentiality	Document access shall be limited according to document type, confidentiality level, workflow role, ownership, and policy.
Auditability	The system shall record all significant document lifecycle events with actor, timestamp, document ID, and version.
Integrity	The system shall preserve approved document versions and prevent

Category	Requirement
Availability	unauthorized modification after approval. Users should be able to access document tasks and status during business operating hours.
Reliability	Workflow state shall remain consistent during notification, signature provider, identity, or storage integration failures.
Retention	Documents and audit records shall follow configured retention and legal hold policies.
Usability	Approvers should be able to understand pending tasks, due dates, and required actions without training-heavy workflows.
Performance	Search, task list, and document status views should respond quickly under normal organizational load.
Scalability	The system should support growth in document volume, workflow count, users, and audit events.
Observability	Administrators shall be able to identify failed notifications, stalled workflows, overdue tasks, storage issues, and e-signature failures.
Compliance Support	The system shall support evidence collection for approval, signature, retention, and access control reviews.
Data Protection	Document contents, metadata, audit records, and signature artifacts shall be protected from unauthorized access or modification.

11. Data Requirements

The system should manage the following data:

1. Document records.
2. Document versions.
3. Document metadata.
4. Document binaries or storage references.
5. Workflow templates.
6. Workflow instances.

7. Approval stages.
8. Approval tasks.
9. Comments and review notes.
10. Approval decisions.
11. Signature requests.
12. Signature status events.
13. Signed document references.
14. User and role references.
15. Notification records.
16. Retention categories.
17. Archival records.
18. Legal hold records.
19. Audit events.
20. Integration request and response records.
21. Access policy records.
22. Document search indexes or derived search data.

Approved document records should preserve the exact version, metadata, approval history, and signature evidence associated with the approval decision.

12. Reporting and Operational Visibility

Authorized users should be able to view:

1. Documents by workflow status.
2. Pending approval tasks.
3. Overdue approval tasks.
4. Approval cycle time.
5. Documents awaiting signature.
6. Failed signature requests.
7. Rejected documents.
8. Documents with requested changes.
9. Documents eligible for archival.
10. Documents under legal hold.
11. Retention deadlines.
12. Workflow bottlenecks.
13. Notification failures.
14. Audit evidence for selected documents.

13. Success Metrics

Metric

Target Direction

Metric	Target Direction
Average approval cycle time	Decrease
Documents with unclear approval status	Decrease
Approval tasks completed before due date	Increase
Documents approved with complete audit trail	Increase
Rework caused by wrong document version	Decrease
Overdue approval tasks	Decrease
Manual email-based approval exceptions	Decrease
Time to find approved document and approval evidence	Decrease
E-signature completion tracking accuracy	Increase
Retention policy compliance	Increase
Unauthorized access incidents	Near zero
Workflow cancellation due to configuration error	Decrease

14. Risks and Open Questions

Area	Question or Risk
Workflow complexity	How complex must routing rules be in the first release?
Approval authority	Are approvers resolved by role, department, named user, monetary threshold, document type, or a combination?
Version policy	Which edits require full re-approval versus limited approval?
Signature requirements	Which document types require external signature?
Signature evidence	What signature artifacts must be stored locally versus referenced from the provider?
Retention	What retention categories and legal hold policies are required?
Access control	Are document permissions based on workflow participation, department, confidentiality level, explicit sharing, or policy rules?
Delegation	Can approvers delegate tasks, and does delegation require audit or administrator approval?
External users	How should external signers or reviewers be authenticated and authorized?

Area	Question or Risk
Storage	Should document binaries be stored internally or in an external storage provider?
Compliance	Which regulatory, contractual, or internal requirements apply to audit logs, signatures, access, and retention?
Notifications	Which reminder and escalation rules are required by default?

15. Release Scope

MVP

The MVP should include:

1. Document upload.
2. Required metadata capture.
3. Basic workflow template configuration.
4. Sequential approval routing.
5. Approval, rejection, and change request actions.
6. Document version history.
7. Approval decision tied to document version.
8. Email notifications for assigned tasks and completion.
9. Basic document search and status visibility.
10. Role-based access control.
11. Audit logging for document, workflow, and approval events.
12. Basic retention category assignment.

Post-MVP

Future releases may include:

1. Parallel approval routing.
2. Conditional workflow rules.
3. External e-signature integration.
4. Signature event processing.
5. Legal hold management.
6. Advanced archival workflows.
7. Delegation and reassignment controls.
8. Advanced escalation rules.
9. External reviewer support.
10. Document storage provider integration.
11. Advanced reporting dashboard.

- 12. Policy-based access control.
- 13. Template-based document generation.
- 14. Compliance evidence export.
- 15. Integration with enterprise document repositories.